

Integrated Approaches to Reliable and Robust CNS in Advanced Air Mobility

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Advanced Air Mobility (AAM) imposes new challenges and requirements on Communication, Navigation, and Surveillance (CNS) systems, particularly in the Very Low Level (VLL) airspace, which is distinct from traditional aviation. While regulations and key system elements for unmanned aerial systems (UAS) and Vertical Take-Off and Landing Aircraft (VTOL) are being introduced, high traffic density, urban dynamics, the shape of the terrain, and other environmental factors will place technological challenges on the AAM ecosystem.

A scalable UTM system is a key element of this ecosystem, with initiatives like CORUS-XUAM advancing standardized flight rules and authorization protocols. Effective UTM operation requires a reliable CNS infrastructure, whereas existing CNS systems from aviation are unable to fully address the specified challenges. As a result, an integrated CNS solution for AAM remains undeveloped.

Considering the requirements of this CNS system, we adapt key elements from existing systems to address the specific needs of AAM and Beyond Visual Line of Sight (BVLOS) operations. For remotely piloted UAS in BVLOS scenarios, the CNS system is divided into two main components, as illustrated in Figure 1: the local CNS system, and the global system associated with the UTM.

Unlike conventional aviation, which primarily relies on voice communication and short digital messages, AAM demands an advanced system capable of managing time-sensitive data flows end-to-end. In addition, navigation systems face significant challenges, such as insufficient accuracy and degraded Non-Line-of-Sight (NLoS) signal propagation, which are particularly problematic for AAM operations due to increased signal interference. Similarly, surveillance methods are limited by high latency and low-resolution visual data, hindering precise flight status updates with accurate real-time situational awareness for AAM operations. These limitations, combined with operational uncertainties underscore the need for precise autonomous flight control to ensure safe and efficient operations.

To address these challenges, a heterogeneous multi-link communication system is recommended to avoid bottlenecks, supported by a resource-aware approach and reliable link health monitoring for optimal performance. To adapt to the locally dynamic propagation environment, influenced by factors such as temperature, moisture, and terrain, fortifying signal propagation through multi-physics coupled simulations is proposed. This approach also enables the integration of novel technologies, such as Reconfigurable Intelligent Surfaces (RIS), to further enhance communication reliability.

The navigation of AAM vehicles in VLL airspace demands high reliability as well, given their proximity to terrestrial infrastructure and the tighter separations between airspace users. However, satellite-based positioning in urban environments faces challenges such as radio interference, shadowing, and multi-path effects, necessitating the use of complementary technologies. Due to advancements in sensor technology, data processing, and sensor fusion, visual-inertial positioning can become a promising approach for ensuring the safe navigation of eVTOLs. Measurements from multiple visual sensors can be combined using Simultaneous Localization and Mapping (SLAM) to augment onboard navigation capabilities. The traditional loop closure methods for drift correction can be adapted using geo-referenced landmarks and geodata-based multi-modal localization.

Digital twin technology addresses the limitations of traditional surveillance systems by creating a virtual replica of each UAS, allowing for real-time simulation and accurate tracking of critical flight metrics, such as position, velocity, and mission status. This technology supports adaptive CNS systems, ensuring timely and accurate operational updates. To achieve a well-mirrored solution, minimizing the discrepancy between the physical system and its digital twin is essential. This error can be modeled through fuzzy randomness and polymorphic uncertainty, accounting for factors such as path deviation, arrival time, and energy consumption. Additionally, digital twins

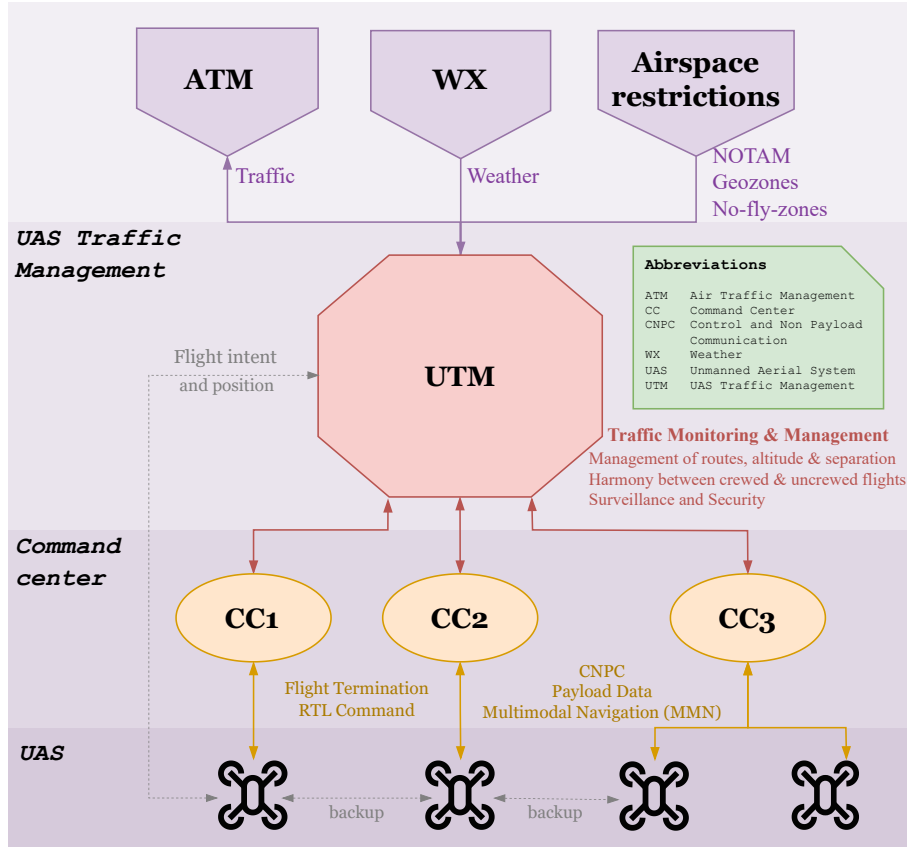


Figure 1: Low level CNS framework for UTM

offer a platform for uncertainty quantification and the development of data-driven solutions.

To address the challenges in CNS for AAM, this work integrates existing and emerging technologies into a unified framework. Our key contributions include:

- An overview of current and emerging CNS technologies, assessing their reliability for safe AAM operations.
- An evaluation of advanced CNS components, such as multi-link communication systems and digital twins, to ensure a cohesive infrastructure.
- Targeted solutions for developing a robust CNS infrastructure that meets UTM requirements.